

5(a)	(coulomb is) ampere second	B1
5(b)(i)	$E = V/d$ or $E = F/Q$	C1
	$F = VQ/d$	A1
	$F = (2.0 \times 10^2 \times 8.0 \times 10^{-19})/4.0 \times 10^{-2} = 4.0 \times 10^{-15} \text{ N}$	
5(b)(ii)	arrow pointing to the left labelled 'electric force' and arrow pointing downwards labelled 'weight'	B1
5(b)(iii)	1. resultant force $= \sqrt{[(3.9 \times 10^{-15})^2 + (4.0 \times 10^{-15})^2]}$	C1
	$= 5.6 \times 10^{-15} \text{ N}$	A1
	2. angle $= \tan^{-1} (3.9 \times 10^{-15}/4.0 \times 10^{-15})$ $= 44^\circ$	A1
5(c)	downward sloping line from (0, 2.0)	M1
	magnitude of gradient of line increases with time and line ends at (T, 0)	A1